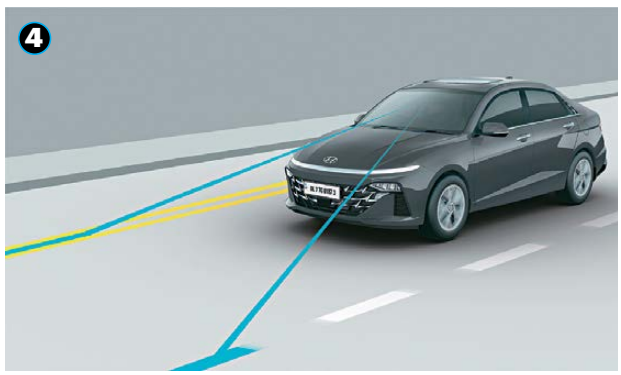




THE MANY FEATURES OF ADAS 1. Lead car departure warning; **2.** Blind Spot Monitor; **3&4.** Lane Follow Assist

In theory, this is a big step forward as not only will these cars provide immediate alerts to drivers in case of transgressions or dangerous situations, like going over the speed limit, there being a car in the blind spot or straying from the lane, but will also take over the braking system of the car and apply emergency brakes if required, not only to avoid a frontal collision but also while reversing out onto the road and in many other situations, depending on the ADAS Level.

However, in practice, neither do all the major systems work flawlessly, nor do they work consistently. The challenges of dealing with mixed and haphazard traffic situations of the sort that exist on Indian roads seem particularly problematic for these technologies.

While some ADAS features rely on external infrastructure—for example, lane-keeping assist will work only with prominent lane markings or the road and the speed limit monitor only when there is proper speed limit signage—many others also need dust-free roads for effective functioning. There have been numerous instances when various ADAS features have gone on the blink when driving on the highway, as the cameras get pasted over by dust and are unable to function accurately.

The biggest challenge comes from the emergency braking systems that violently apply the brakes if the system concludes that there is an impending collision even though the driver is alert and in control and knows that there is not the minutest chance of an accident.

I have had many experiences like this. The guard slowly opening the gate to the compound—causing the vehicle to slam on the brakes. Many other times, it is when pedestrians are crossing the road and both the pedestrian and the driver are aware of the situation. The pedestrian will be across the road by the time the vehicle gets there but the system concludes otherwise and slams the brakes sparking a chain collision as multiple cars rear-end each other. Other instances are when one has to reverse a parked vehicle onto the road by gently easing it out and the vehicle cuts power and brakes leading to a hopeless situation where it becomes impossible to get out of the parking slot. And then one has to switch off this particular emergency braking system and then remember that either it may remain switched off forever or come on again when the car is restarted. For most people I know, the endeavour is to keep such systems switched off.

Instead of spending enormous amounts of money to load cars with these complex ADAS equipment, perhaps it would be a better idea to invest in shoring up the infrastructure so that these systems can work well in the first place and in training road users to bring order on the roads. Not only will that help prevent accidents, it will also help ADAS to work when it is actually required. ■